

FINE TUNING REAL WOLD EXAMPLES

Introduction to Fine-Tuning

Fine-tuning is a process in which a pre-trained Large Language Model is further trained using domain-specific data so that it can understand and generate outputs according to a particular business requirement or use case. Instead of learning from general internet data, the model learns from structured and repeated patterns such as company standards, domain-specific language, and expected response formats. This helps the model produce more accurate, consistent, and context-aware outputs. Fine-tuning is especially useful when we need the model to follow strict rules, generate structured responses, or replicate a particular style repeatedly. However, it is important to note that fine-tuning is not suitable for real-time or frequently changing data, and in such cases, other techniques like APIs or Retrieval-Augmented Generation are preferred.

Automation Code Generation (Framework-Based Fine-Tuning)

In automation code generation, fine-tuning is used to train the model on a company's internal framework standards, including coding style, naming conventions, reusable methods, and test case structures. By learning from this structured dataset, the model becomes capable of generating code that aligns with the organization's development practices. For example, if a company follows the Page Object Model in Selenium automation, the fine-tuned model will automatically generate test scripts that include proper class structures, method reuse, and standardized naming patterns. This reduces manual effort, improves productivity, and ensures consistency across the codebase. In real-time development environments, this approach helps teams maintain high-quality and uniform automation scripts without requiring developers to repeatedly write boilerplate code.

Zepto (Multilingual Spell Correction)

Zepto implemented a multilingual spell correction system using advanced techniques like prompt engineering, few-shot learning, structured JSON outputs, and Retrieval-Augmented Generation rather than traditional fine-tuning. Although they refer to it as "instruct fine-tuning," the model weights were not actually modified, which means it is not classical fine-tuning. Instead, the system works by providing the model with contextual examples and structured instructions so that it can correct misspelled words across different languages. For instance, when a user enters a misspelled query, the system uses retrieved examples and predefined prompts to guide the model in generating the correct spelling. This approach is highly effective for handling dynamic and multilingual input data without requiring costly retraining of the model. It highlights that in many real-world applications, prompt engineering combined with RAG can be more practical than full fine-tuning.

Ola (Krutrim Platform)

Ola's Krutrim platform demonstrates a comprehensive ecosystem for working with large language models, including foundation models, instruction-tuned models, and fine-tuning infrastructure. This platform allows developers and organizations to customize models according to their specific use cases by training them on domain-specific datasets. For example, Ola can fine-tune models for regional language understanding, customer support automation, or transportation-related queries. By providing infrastructure for fine-tuning, Krutrim enables businesses to build AI systems that are more aligned with their operational requirements. This approach is particularly useful in industries

where localized knowledge and domain expertise are critical for delivering accurate and meaningful responses.

Uber Eats (Optimizing LLM Training)

Uber Eats uses a combination of open-source models and in-house techniques to optimize large language model training for large-scale applications. Their approach focuses on improving performance, scalability, and efficiency in handling massive amounts of data. Fine-tuning plays a role in adapting models to specific business needs such as food recommendations, delivery optimization, and customer interactions. By customizing models with relevant datasets, Uber ensures that the outputs are aligned with user preferences and operational requirements. This helps in delivering personalized and efficient services while maintaining high performance across a global platform.

Netflix (Personalization Applications)

Netflix uses foundation models and fine-tuning techniques to enhance personalization in its platform. The goal is to improve recommendation systems by understanding user behavior, preferences, and viewing patterns. By training models on domain-specific data such as watch history and content metadata, Netflix ensures that its recommendations are highly relevant and engaging. Fine-tuning helps the model adapt to the entertainment domain, where understanding user taste is crucial. This leads to better user experience, increased engagement, and improved retention rates, as users are more likely to find content that matches their interests.

Grammarly (Writing Assistance)

Grammarly uses fine-tuned language models to provide advanced writing assistance features such as grammar correction, paraphrasing, text simplification, and multilingual support. By training the model on linguistic data and writing patterns, Grammarly ensures that it can understand context and provide accurate suggestions. The fine-tuned model is capable of adapting to different writing styles and tones, making it useful for both professional and casual communication. This application demonstrates how fine-tuning can enhance the quality of generated text and improve user productivity in writing tasks.

Spotify (Personalized User Experience)

Spotify uses AI and machine learning techniques, including LLM-based approaches, to create personalized user experiences such as music recommendations and “Wrapped” summaries. Fine-tuning helps the model understand user preferences, listening habits, and trends, enabling it to generate insights that are tailored to individual users. By analyzing large amounts of user data, Spotify can deliver highly engaging and customized content. This improves user satisfaction and strengthens user engagement by providing a unique and personalized experience for each listener.

Pinterest (Content Understanding and Recommendation)

Pinterest explores fine-tuning and domain-specific model adaptation to improve content understanding and recommendation systems. The platform deals with a vast amount of visual and textual data, and fine-tuning helps the model better interpret this information. By training models on

platform-specific data, Pinterest can enhance its ability to recommend relevant content to users based on their interests and behavior. This leads to more accurate and meaningful recommendations, improving the overall user experience.

Swiggy (Customer Support Systems)

Swiggy explored fine-tuning for its customer support systems but found that prompt engineering and AI agents are more practical for handling real-time interactions. Customer support often involves dynamic data such as order status, delivery updates, and refunds, which change frequently. In such cases, fine-tuning is not ideal because it cannot handle real-time updates effectively. Instead, Swiggy uses APIs, agents, and retrieval-based systems to fetch live data and provide accurate responses. This example highlights an important limitation of fine-tuning and emphasizes the need to choose the right approach based on the problem type.